Rahul Aditya Butta

Machine Learning Engineer: LLMs, NLP, computer vision, speech, classical ML | Bangalore, India
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Experience

Zynix AI *(healthcare AI for value-based care)* Sep 2025 – Present
Machine Learning Engineer *Bangalore, India*

- Built 30-day admission, ER and readmission models (XGBoost) on 1.18M member-months of Medicare claims in Databricks: AUROC up to 0.79 and 4.5× AUPRC lift on a patient-disjoint holdout.
- Replaced Youden cut-offs with thresholds sized to care-team capacity: 2.4× higher 30-day precision on an out-of-time fold, with weekly outreach flags cut from about 1,600 to 300.
- Built, alone, a fraud and waste detection system on Databricks for medical-equipment claims (6,500+ suppliers): policy rules, anomaly ensemble, graph features, positive-unlabeled learning.
- Validated it with no labels: all 5 suppliers the HHS-OIG later excluded were already in its top-0.53% risk tier, and held-out confirmed fraud suppliers ranked in the top 3% (median rank 1).
- Trained a Qwen3-4B claims foundation model with LoRA on 4× L40S and 4× RTX 4090 nodes, fixing a data-parallel gradient-sync bug; it beat XGBoost on 12-month CKD progression, AUROC 0.674 vs 0.645.
- Designed the modular, multi-client architecture of the in-house LangGraph calling agent: per-client call flows built from YAML config, a graph-agnostic LLM service and plug-in memory extractors.
- Added streaming speech enhancement (fine-tuned DeepFilterNet) and two-stage voicemail detection to the agent.

Carnot Research *(IIT Delhi)* Mar 2025 – Sep 2025
Machine Learning Engineer *Hauz Khas, New Delhi*

- Productionised GFM-RAG, graph-neural retrieval over a document knowledge graph, for 12 Indian languages: macro Recall@10 rose from 59.8% to 77.3% and multi-hop success from 52% to 72% vs ColBERTv2.
- Built an on-device voice and vision agent on Jetson AGX Orin: faster-whisper, LLaVA-1.5 7B quantised to 4-bit with AWQ, Chatterbox TTS, and an attention-sink KV cache for long sessions.
- QLoRA-fine-tuned Qwen 7B for the citizen chatbot in Sachet, India's national disaster-alert app.

Videoverse Aug 2024 – Nov 2024
Machine Learning Engineering Intern *Remote*

- Fine-tuned VideoMAE for live cricket event detection in a production ensemble serving Disney+ Hotstar and other streaming platforms; prioritised on 2 event classes, where it beat the VLMs (∼74% precision).

Dronapay Apr 2024 – Jul 2024
Data Science Intern *Remote*

- Raised fraud-detection accuracy by 12 points with keystroke-dynamics features.
- LoRA-fine-tuned Llama 2 7B with retrieval to turn natural-language fraud rules into executable JSONLogic: 82% logical and 74% execution accuracy.

Open-Source Projects

- **fused-rmsnorm-triton**: fused residual-add + RMSNorm Triton kernel, forward and backward. Runs at 85–89% of peak memory bandwidth on an RTX 4060, 8–12× faster than eager PyTorch, on par with torch.compile.
- **Attention_sinks**: attention-sink KV cache for Transformers 5 with fused Triton eviction and CUDA-graph decode. Perplexity within 6% of a full cache to 40K tokens at constant memory; found and fixed a bf16 drift bug.

Publication

Boddu, **Butta** et al. *QueryAssist: Multimodal Verbal Specifications to Structured Query Conversion Model Using Word Vector-Based Semantic Analysis*. InCITe 2024, Springer. **Best Paper Award.**

Education

Indian Institute of Information Technology, Pune, B.Tech, Computer Science and Engineering 2020 – 2024

Skills

Classical ML: feature engineering, classification, XGBoost, scikit-learn, anomaly detection, PU learning, calibration
Deep learning: PyTorch, TensorFlow/Keras, CNNs, RNNs, Transformers, Hugging Face, LoRA, LangGraph
Systems: Triton, CUDA graphs, data-parallel and 3D-parallel training (Picotron), AWQ, NVIDIA Jetson
Data: Databricks (SQL, Delta Lake, Workflows), Neo4j, GraphRAG, Python, pandas